A Single Modular Character Is Complete for SYM∘LTF Counting

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Abstract

Williams' algorithmic method turns nontrivial algorithms estimating acceptance probabilities into circuit lower bounds, and the strongest known gate-count lower bounds for depth-two threshold circuits follow this route. For circuits $C(x) = f(\sum_{i=1}^{q} T_i(x))$ with an arbitrary symmetric top $f : \{0, \dots, q\} \to \{0, 1\}$ and linear threshold bottom gates T_i on h variables, a natural plan is to estimate a single modular character of the bottom-gate sum and recover f by linearity. We show that this primitive is as hard as the general problem. Let R be the least power of two larger than q and $\omega = e^{2\pi i/R}$. In deterministic polynomial time we construct $M = O(q^2)$ linear threshold functions $U_1, \dots, U_M$ on $h + 2\log_2 R + 1$ variables, appearing with unit coefficients in the character exponent (their internal threshold weights remain general), such that $\mathbb{E}\,\omega^{\sum_j U_j} = R^{-1} \Pr_x[C(x) = 1]$. The compression comes from logarithmically many uniform selector variables, combining character orthogonality with exact cancellation of rejected top values. Conversely, the character expectation is a Fourier combination of R residue probabilities, each a symmetric-top counting query, so the primitives are polynomially interreducible when R and M are polynomially bounded. Estimating the acceptance probability within ε requires the character expectation within ε/R, and in the reverse direction accuracy ε/R per residue suffices. Deterministic algorithms transfer to deterministic algorithms; randomized ones transfer with a union bound over at most R queries. The character expectation is complex-valued and differs from the Boolean acceptance probability of a single MOD_R gate. The equivalence is polynomial and need not preserve a particular gate-size exponent or precision regime. The result is a representation theorem: it gives no new algorithm or circuit lower bound and makes no claim about characters with a fixed modulus.

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Problem and contribution

Williams' algorithmic method turns a nontrivial algorithm estimating the acceptance probability of a circuit class into a lower bound for a function in E^{NP} [1]. The strongest known gate-count lower bounds for depth-two threshold circuits are obtained by this route [2]; not all such lower bounds are: for example, the earlier bounds of Kane and Williams [3] use random restrictions instead, as [2] itself points out. For circuits with a symmetric top gate over linear threshold functions (LTFs), one natural plan is to estimate a single modular character of the bottom-gate sum and recover the symmetric top by linearity. We show that this primitive is not easier than the general problem.

Main result (Theorem 1): counting for $\mathrm{SYM}_q \circ \mathrm{LTF}_h$ reduces exactly, with $O(q^2)$ LTFs and $2\log_2 R + 1$ added selector variables, to one character expectation $\mathbb{E}\,\omega^{\sum_j U_j}$ in which every LTF output appears with unit coefficient in the exponent (the internal threshold weights of the U_j remain general) with $\omega = e^{2\pi i/R}$, R the least power of two larger than q. Section 2 gives the converse reduction, so the two counting primitives are polynomially equivalent.

The supplement contains a numerical check of identity (1) on small random instances (floating point, tolerance 10^{-9}); it is not part of the proof.

Not claimed: no new algorithm, no circuit lower bound, and no statement about fixed-modulus characters.

1. The representation theorem

An LTF is written as $T(x) = 1[L(x) \geq \theta]$, with rational weights. Clearing denominators changes the encoding length only polynomially.

Theorem 1 (single-character completeness with unit exponent coefficients)

Let $q \geq 1$, let $T_1, \dots, T_q \in \mathrm{LTF}_h$, and let $f : \{0, \dots, q\} \to \{0, 1\}$. Put $R = 2^\ell > q$, where R is the least such power of two, and let $\omega = e^{2\pi i/R}$. There is a polynomial-time construction of

$$U_1, \dots, U_M \in \mathrm{LTF}_{h+2\ell+1}$$

with

$$M = O(qR + \ell^2 R + R^2) = O(q^2)$$

such that

$$\mathbb{E}_{x,a,s,u}\left[\omega^{\sum_{j=1}^{M} U_j(x,a,s,u)}\right] = \frac{1}{R}\Pr_x\left[f\left(\sum_{i=1}^{q} T_i(x)\right) = 1\right]. \tag{1}$$

Here $a, s \in \{0,1\}^\ell \cong \mathbb{Z}_R$, $u \in \{0,1\}$, and all variables are uniform and independent. An estimate of the left-hand side with complex error η recovers the acceptance probability with error at most $R\eta < 2(q+1)\eta$. Equivalently, to estimate the acceptance probability within ε it suffices to estimate the character expectation within ε/R.

Consequently, $\mathrm{SYM}_q \circ \mathrm{LTF}_h$-CAPP many-one reduces, with polynomial size blow-up, to estimating one character expectation $\mathbb{E}\,\omega^{\sum_j U_j}$ of LTFs on $h + O(\log q)$ variables with unit exponent coefficients. This is a complex-valued expectation; it is not the Boolean acceptance probability of a single MOD_R gate, which would be one of the residue probabilities in Section 2.

Proof

Write $S(x) = \sum_{i=1}^{q} T_i(x) \in \{0, \dots, q\}$, and extend f by zero to $\mathbb{Z}_R$. Define

$$\Phi(x, a, s, u) = a(S(x) - s) + \frac{R}{2}u(1 - f(s)) \pmod R. \tag{2}$$

For fixed x, a, s, averaging over u gives

$$\mathbb{E}_u\,\omega^{\Phi} = \begin{cases} \omega^{a(S-s)}, & f(s) = 1, \\ \frac{1}{2}(1 + \omega^{R/2})\omega^{a(S-s)} = 0, & f(s) = 0. \end{cases} \tag{3}$$

Character orthogonality gives

$$\mathbb{E}_a\,\omega^{a(S-s)} = 1[S = s], \tag{4}$$

because $R > S$. Averaging next over uniform s proves

$$\mathbb{E}_{a,s,u}\omega^{\Phi(x,a,s,u)} = R^{-1} f(S(x)). \tag{5}$$

It remains to realize Φ as a sum of LTF outputs, each with coefficient one. Write $a = \sum_{b<\ell} 2^b a_b$ and $s = \sum_{c<\ell} 2^c s_c$. For each bottom gate choose an integer $B_i > |\theta_i| + \sum_j |w_{ij}|$. Then

$$a_b \wedge T_i(x) = 1[L_i(x) + B_i a_b \geq \theta_i + B_i] \tag{6}$$

is an LTF. Replicating it 2^b times realizes

$$aS = \sum_{i=1}^{q} \sum_{b<\ell} 2^b (a_b \wedge T_i(x)), \tag{7}$$

using $q(R-1)$ copies.

For the term as, retain only products $a_b s_c$ with $b + c < \ell$, since all other terms are multiples of R. Modulo R,

$$-2^{b+c} a_b s_c \equiv 2^{b+c}(1 - a_b s_c) - 2^{b+c}, \tag{8}$$

and $1 - a_b s_c = 1[a_b + s_c \leq 1]$ is an LTF. Replication and constant-one LTFs implement the entire $-as$ term with $O(\ell^2 R + R)$ gates.

For each $t \in \mathbb{Z}_R$, the singleton selector

$$V_t(s,u) = 1[(s,u) = (t,1)] \tag{9}$$

is an LTF. If t_c is the c-th bit of t, set

$$d_t(s) = \sum_{c:t_c=1} s_c + \sum_{c:t_c=0} (1 - s_c), \qquad V_t(s,u) = 1[d_t(s) + u \geq \ell + 1]. \tag{10}$$

For each t with $f(t) = 0$, replicate V_t $R/2$ times. This realizes the final term in (2) using at most $R^2/2$ copies. Thus the total number of LTFs is $O(qR + \ell^2 R + R^2) = O(q^2)$, proving (1). □

2. Reverse direction and exact scope

Given $U_1, \ldots, U_M$, each residue probability

$$\Pr\left[\sum_j U_j \equiv t \pmod R\right]$$

is a symmetric-top CAPP query. The character expectation is their discrete Fourier linear combination $\sum_t \omega^t \Pr[\sum_j U_j \equiv t]$; by the triangle inequality, estimating each of the R residue probabilities within ε/R estimates the character expectation within ε. Hence, when R and M are polynomially bounded, the two counting primitives are polynomially interreducible.

Precision, size and determinism. Both reductions are deterministic polynomial-time transformations; the forward direction makes one query and the reverse direction makes R queries. To estimate the original acceptance probability within ε, a character algorithm must handle $h + 2\log_2 R + 1$ variables, $O(q^2)$ gates and additive error ε/R. Polynomial equivalence therefore does not preserve a particular gate-size exponent, and an algorithm valid only for a restricted size or precision regime need not transfer. A deterministic estimation algorithm, as required in the applications of the algorithmic method [1,2], yields a deterministic algorithm on the other side. A randomized algorithm with success probability at least $1 - \delta$ transfers with success probability $1 - \delta$ in the forward direction and at least $1 - R\delta$ in the reverse direction, by a union bound.

The construction is deliberately selector-based. A single fixed original-variable frequency $\mathbb{E}_x \omega^{kS(x)}$ supplies two real numbers, while $\mathbb{E} f(S(x)) = \sum_t f(t) \Pr[S(x) = t]$ ranges over a $(q+1)$-dimensional space of profiles f; hence for $q \geq 3$ no fixed frequency of the original sum determines $\mathbb{E} f(S(x))$ for every f and every distribution of S. The selectors a, s, u, not an identity for one original frequency, provide the compression.

The growing modulus is used essentially: $R > q$ is what makes (4) an exact indicator of $S = s$. We do not know a comparable single-character construction with a fixed modulus, and we make no claim either way for that model.

3. Consequence for depth-two threshold algorithms

Williams' algorithmic method turns nontrivial acceptance-probability estimation (CAPP) algorithms into circuit lower bounds [1], and the algorithmic progress on THR ∘ THR in [2,4,5] follows this route. Any method that first builds a general algorithm for all characters of this kind and then recovers arbitrary symmetric tops by linearity does not work with a strictly easier subproblem: the same primitive already represents arbitrary SYM ∘ LTF-CAPP after polynomial blow-up.

This is narrower than a CAPP lower bound. It leaves open algorithms that exploit the structure of a specific symmetric profile, including cancellations invisible frequency by frequency. It also leaves open models with a fixed modulus, a different query model, or a different gate family.

4. Relation to prior work and limitations

The result is a representation theorem. Its ingredients, character orthogonality and the realization of products of bits by threshold gates, are standard; we claim only the specific consequence that one character with unit exponent coefficients and $O(\log q)$ selector variables already carries arbitrary symmetric-top counting. Related settings include the depth-two satisfiability, counting and lower-bound algorithms of Williams [1], Tamaki [4], Alman, Chan and Williams [5]

and Chen, Tal and Wang [2], which use batch evaluation and probabilistic-polynomial representations of threshold functions, and the pseudorandom generators for constant-depth threshold circuits by Hatami, Hoza, Tal and Tell [6], which use circuit simplification. The closest Fourier precedent is Gopalan, Kane and Meka [7], who relate fooling Fourier shapes to fooling modular tests and combinatorial shapes (Sections 1.1 and 9 of [7]). There each factor depends on a separate block of input coordinates, whereas here the LTFs overlap arbitrarily and the auxiliary selectors combine the desired symmetric profile into one expectation. In a focused comparison with these works we did not find the single-character reduction above; this is a bounded comparison, not an exhaustive survey. The ingredients are standard and credited as such; the contribution is their combination with the stated gate and variable bounds.

The theorem does not prove a new lower bound against THR $\circ$ THR, does not imply $P \neq NP$, and does not show that any particular symmetric profile is hard. Its precise contribution is to remove one generic shortcut: growing-modulus single-character estimation with unit exponent coefficients is already complete for general symmetric-top counting when selector variables are allowed.

References

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